

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 3– Mock Interview

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 3–Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	T.Nikhitha	Reg. No.:	192424114
Section / Batch:	2024-2028/AI&DS	Date:	27/08/2026

Code of Conduct

I T.Nikhitha-192424114 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T.Nikhitha

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5
Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6
Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8
Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9
Integrated Case Study (Analytical & Problem Solving)	Real-world scenario involving discourse analysis, conversational agents, language generation, and machine translation	Q10

1. Natural language processing is a branch of Artificial intelligence that enables computers to understand, interpret, and generate human language. It combines linguistics, computer science, and machine learning to process text and speech.
2. The main goals of NLP are to help computers understand, interpret, and generate human language accurately. It also aims to improve communication between humans and computers through natural language.
3. NLP is used in machine translation, chatbots, speech recognition, sentiment analysis, and text summarization. It is also used in spam filtering, question answering, and search engines.
4. NLP is the overall field that deals with processing human language. NLU focuses on understanding the meaning of text, while NLG focuses on producing human-like text.
5. The major challenges include ambiguity, understanding context, handling multiple languages, and processing informal language. Sarcasm, idioms, etc. also make NLP difficult.
6. Human language is complex because words and sentences often have multiple meanings. Understanding context, emotions & cultural expressions makes language processing challenging for computers.
7. Ambiguity occurs when a word, phrase, or sentence has more than one possible meaning. NLP systems must use context to determine the correct interpretation.

8. Lexical ambiguity occurs when a word has multiple meanings. Syntactic ambiguity arises from different sentence structures, semantic ambiguity from multiple interpretations, depends on context.
9. A language model predicts the probability of a sequence of words. It is used in applications like text prediction, & machine translation.
10. Rule-based NLP relies on manually created grammar and language rules. Statistical NLP learns language patterns from large datasets using probability and machine learning techniques.
11. A Regular expression is a sequence of characters that defines a search pattern. It is commonly used for text searching, validation & pattern matching.
12. Common regular expression symbols include ., *, +, ?, ^, \$, [], and (). These help define search patterns for text processing.
13. The (*) symbol means "zero or more occurrences" of the preceding character or pattern. For ex: ab^* matches ab , abb & $abbb$.
14. The + symbol means "one or more occurrence" of the preceding character. For ex: ab^+ matches ab , abb , $abbb$ but not a .
15. The * operator allows zero or more occurrences, while the + operator requires at least one occurrence. $\therefore$ * is more flexible ^{than} +.
16. Character classes allow matching any one character from a specified group. For example $[abc]$ matches either a , b or c .
17. A finite state automation is a mathematical model used to recognize patterns in strings. It consists of states connected by transition.

18. An FSA consists of a finite set of states, transitions, a start state and one or more final states. These define input string is accepted.
19. FSAs are used in NLP for tokenization, lexical analysis & morphological analysis. They efficiently recognise valid word patterns.
20. In a Deterministic Finite Automaton, each input symbol has only one possible transition from a state. In a Non(FA), are allowed.
21. Morphology is study of the internal structure of words and how they are formed from smaller meaningful units called morphemes.
22. A morpheme is the smallest meaningful unit. It can be a complete word such as a prefix or suffix, ex: -unhappy = "un" + "happy".
23. A free morpheme can stand alone as a independent word, such as book or run. A bound morpheme cannot stand alone must be attached to another morpheme.
24. Inflectional morphology modifies a word to express grammatical information such as tense, numbers, genders, its basic meaning will change.
25. Derivational morphology forms a new word by adding prefixes or suffixes, often changing the meaning.
26. Inflectional morphology changes the grammatical form of a word without changing its meaning. Derivational morphology creates a new word and may change both its meaning and grammatical category.
27. An example is walk - walked, where the suffix -ed indicates the past tense. Another example is dog -> dogs, where -s indicates the plural form.

28. An example is kind $\rightarrow$ kindness, where the suffix -ness creates a noun from an adjective. Another ex: happy $\rightarrow$ unhappy.
29. Morphological parsing is process of analyzing a word into its constituent morpheme, such as roots, prefixes, and suffixes.
30. Finite-state morphological parsing uses finite-state automata to analyze & recognize valid word forms efficiently. It is used in spell checking & text analysis.
31. Stemming is process of reducing words to their root prefixes. It uses simple rules & may not always produce a valid dictionary word.
32. Lemmatization is the process of converting a word into its base or dictionary form, called the lemma. It considers the word's context and grammatical structure to produce meaningful root words.
33. Stemming removes affixes using simple rules and may generate non-dictionary words. Lemmatization uses vocabulary making it more accurate.
34. The porter stemmer is one of the most popular rule-based stemming algorithms used in NLP. It removes common English suffixes to obtain the stem of a word.
35. The porter stemmer may produce incorrect by removing too many. It does not consider the context of words, which can reduce accuracy.

36. An N-gram is a sequence of N consecutive words in text. It is commonly used in language modeling, text prediction.
37. A unigram model treats each word independently and predicts words based only on their individual probabilities.
38. A bigram model predicts a word based on the immediately preceding word. It captures simple contextual relationship.
39. A trigram model predicts a word based on the two previous words in a sentence. It captures more context than unigram and bigram models, resulting better predictions.
40. Trigram considers three consecutive words. Higher-order-N-grams capture more context but require more data.
41. The probability of a bigram is calculated as $P(w_2|w_1) = \text{Count}(w_1, w_2) / \text{Count}(w_1)$. It represents the likelihood of a word occurring after the previous word.
42. N-gram models capture local word sequences and help predict the next word in a sentence.
43. N-grams that appear in the training data. Any unseen N-gram receives a probability of zero.
44. Even if they are valid in real language. This causes poor predictions and reduces the effectiveness of language
45. Smoothing is a technique that assigns small probabilities to unseen N-grams instead of zero. It improves the robustness and accuracy of language models.

46. unseen word sequences and improves prediction accuracy. It allows language models to handle new or rare combinations of words effectively.
47. N-gram counts are unavailable, for ex:- if a trigram is unseen, the model falls back to bigram or unigram.
48. unigram, bigram and trigram models using weighted averages. It provides more reliable predictions than relying on a single n-gram.
49. Lower entropy indicates better predictions, while higher entropy indicates greater uncertainty.
50. model is uncertain about predicting the next word. Accuracy and less confidence in its output.
51. These categories include nouns, verbs, adjectives, adverbs, and others. It is one of the basic tasks in Natural LP.
52. POS tagging helps computers understand the grammatical structure of a sentence.
53. The major English word classes are nouns, pronouns, verbs, adjectives, & interjections. These classes define the words.
54. A noun is a word that names a person, place, thing. Ex:- include student, city, computer, & freedom.
55. A pronoun is word used in place of a noun to avoid repetition. Ex:- include he, she, it, they and we...
56. A verb is word that expresses an action, occurrences or state of being.

57. An adjective is word that describes or modifies a noun or pronoun. Ex: beautiful, tall, red & three.
58. An adverb modifies a verb, adjectives or another adverb. It tells how, when, where something happens.
59. A determiner is word placed before a noun to specify it. It include a, an, the, this and many, They provide information about quantity or possession.
60. A preposition is a word that shows the relationship between a noun and another word in a sentence.
61. A conjunction is a word that joins words, phrases. include and, but, or etc. They help connect ideas within a sentence.
62. A POS tagset is a collection of standardized tags used to represent different parts of speech. Ex: penn treebank Tagset.
63. POS ambiguity occurs when a word can belong to more than one grammatical category. Ex: -book. correct tag is determined using context.
64. Rule-based POS tagging assigns tags using manually created grammar and linguistic rules. It considers word endings, sentence structure, and context.
65. Stochastic POS tagging uses statistical methods and probability to assign POS tags. It is trained on annotated text.
66. Transformation-based POS tagging combines rule-based and statistical approaches.

67. Rule-based POS tagging relies on manually written grammar rules, while stochastic POS tagging uses probabilities learned from training data.
68. Many English words can perform different grammatical functions depending on the sentence. Ex:- book is a noun.
69. The POS of book is identified by analyzing its role in sentence. "I read a book", "Book a room".
70. Context helps determine the correct grammatical category of words with multiple meanings.
71. A CFG is a formal grammar used to describe the syntactic structure of sentences.
72. A CFG has 4 components: Terminals, Non-terminals, productions rules, start symbol. These define how valid sentences are formed.
73. A terminal is a symbol that appears in final sentence example: The, boy, runs.
74. A Non-terminal represents a grammatical category that can be expanded examples: S, NP, VP.
75. A production rule defines how a non-terminal can be replaced by other symbols. Ex:- $S \rightarrow NP VP$.
76. A parse tree shows the grammatical structure of a sentence. It represents how words are grouped into phrases.
77. A parse tree shows the complete grammar derivation of a structure.

It means a sentence consists of a noun phrase and a verb phrase.
Example: "The boy runs".

Syntactic analysis identifies the grammatical structure of a sentence. It determines how words & phrases are related.

80. Parsing is process of analyzing a sentence using grammatical rules.
It produces the syntactic structure of the sentence.

81. Top-down parsing starts with the start symbol and works toward the input words.

82. Bottom-up parsing starts with the input words & builds large structures.

83. Top-down parsing moves from grammar to words.
Bottom-up parsing moves from words to grammar.

84. An Earley parser is an Algorithm used to parse context-free Grammar. It can handle ambiguous & complex grammar structures.

85. It can handle complex & ambiguous Natural language sentences.

86. Grammatical agreement means related words must match the grammatical features. These include Number, Person & gender.

87. The subject & verb must agree in number & person.

Ex:- She runs, but they run.

88. Sub categorization specifies what types of complements a word or verb can take. Ex: "give" usually takes an object & a recipient.

89. A feature structure represents linguistic information using attribute-value pairs.

90. A probabilistic context-free Grammar assigns a probabilities to each grammar rule.

91. Discourse refers to connected sentences or conversations.
92. Discourse analysis studies the meaning & structure of the connected text or conversations.
93. Reference resolution identifies the entity referred to a word or phrase. Example :- "Ravi lost his book" $\rightarrow$ his = Ravi
94. Anaphora resolution identifies what a pronoun refers to in previous text.
95. Coherence is the logical connection b/w ideas in a text. It makes the text easy to understand as a whole.
96. Discourse structure describes how sentences & ideas are organized & connected.
97. Coherence refers to the logical flow of ideas; cohesion refers to grammatical links b/w sentences, such as pronouns & conjunctions.
98. A dialogue system is a computer system that communicates with users using natural language.
99. A conversational agent is an AI system that interacts with users through natural language.
100. A dialogue act represents the purpose of an utterance in a conversation. Ex:- include question, ^{answer} ~~ans~~, request & confirmation.